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Explainable IDS Under Adversarial Pressure

NSL-KDD intrusion detection with interpretable decisions, explanation stability, adversarial attacks, and tested defenses.

Final model: Adv+ExtraTrees Ensemble IDS

0.9063
Binary F1

0.9626
PR-AUC

One sentence to defend

The final IDS gives the best tested balance between clean detection, rare-family recall, explainability, stability, and adversarial robustness.

Key result: transfer-PGD evasion drops to 0.50% at eps 0.03 and 0.00% at eps 0.06, 0.10, and 0.15.

From Dataset To Defensible IDS Decision

125,973
KDDTrain+ records

22,544
KDDTest+ records

478
Preprocessed features

Goal: normal vs attack detection, then family recall for DoS, Probe, R2L and U2R.

Preprocess
One-hot, scaling, traffic features

Train
LR, RF, ET, XGB, Torch MLPs

Explain
TreeSHAP, IG, SmoothIG

Attack
SHAP/IG evasion, PGD

Defend
Adv. training, robust trees

IDS task
Binary decision detects normal vs attack traffic. Family recall checks whether difficult classes are hidden by global metrics.

Evaluation logic
We report F1, PR-AUC, balanced accuracy, stability, evasion rates, and defense reductions instead of accuracy alone.

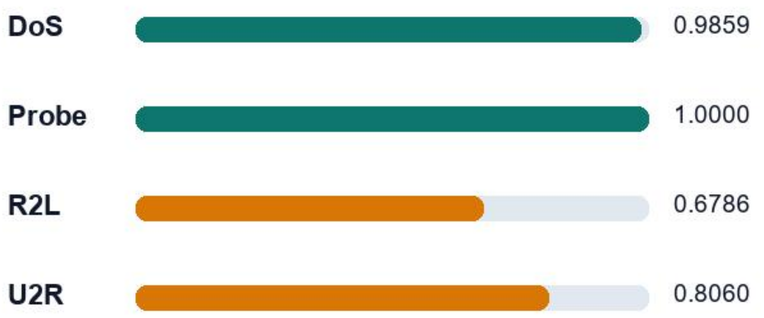
Reproducibility
The CUDA pipeline fixes seeds, saves figures and summary.json, and documents heavy options for robust XAI runs.

Best Default Model: Adv+ExtraTrees Ensemble

Candidate comparison

Model	F1	PR-AUC	Bal. Acc.
Logistic Regression	0.8344	0.9388	0.8442
Torch MLP CUDA	0.8470	0.9077	0.8605
SHAP-Robust ExtraTrees	0.8912	0.9592	0.8941
Adv+ExtraTrees	0.9063	0.9626	0.9064
Security-OR	0.8796	0.9451	0.8839

Family recall of final IDS



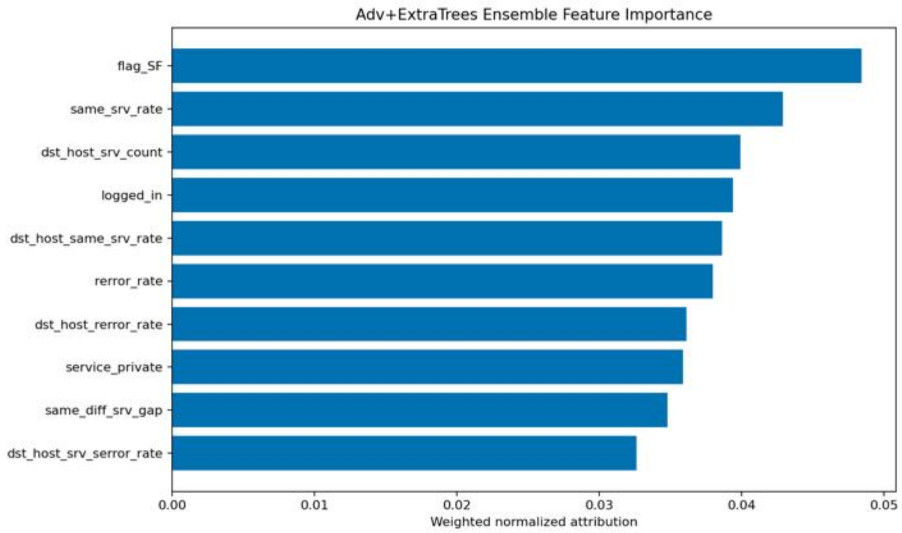
Deduction: the selected model is the best operational compromise: strong clean metrics, better rare-family recall, and strong adversarial behavior.

Explanations Show Meaningful Network Evidence

How to read the figure

The ranking shows which features most influence the final IDS decision. A value such as 0.04 is interpreted relative to the same explanation method, not across all explainers.

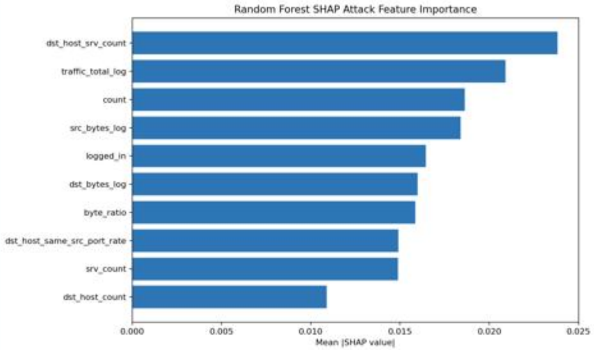
Important: SHAP and SmoothIG use different units. Compare feature order, security meaning, and stability, not raw bar heights.



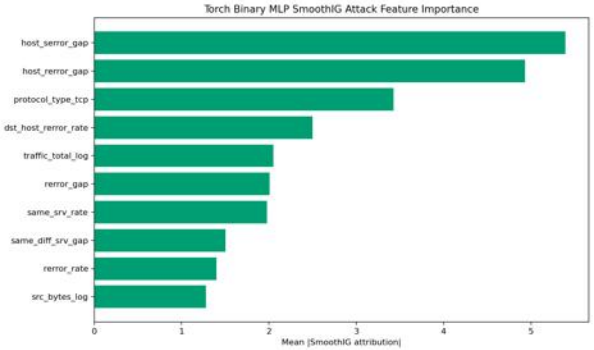
Final ensemble: connection status, service concentration, login state, and error-rate features are logical IDS evidence.

Interpretation: the model relies on security-relevant behavior, not arbitrary variables.

Different Explainers, Same Security Story



TreeSHAP explains tree models through feature contributions.



SmoothIG explains neural scores with smoothed gradient

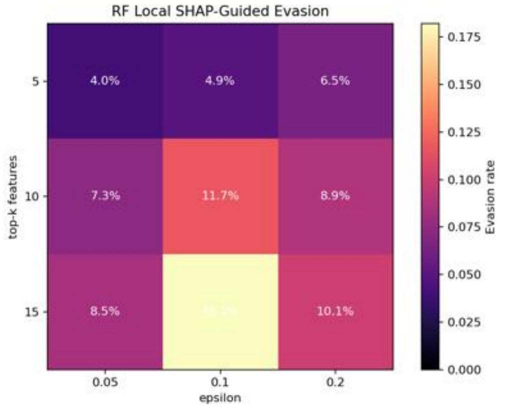
Stability results

- RF SHAP local Jaccard: 0.880
- Torch IG local Jaccard: 0.881
- SmoothIG local Jaccard: 0.899
- Final ensemble local Jaccard: 0.845

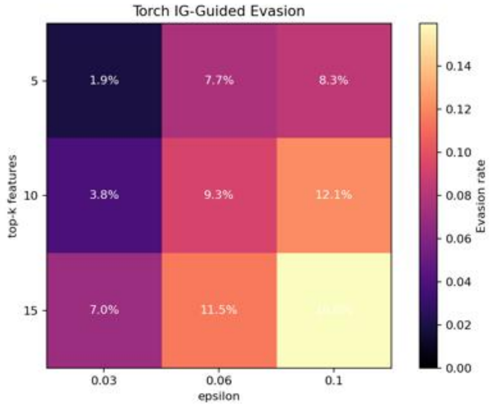
Stable explanations support analyst trust, but they also reveal consistent attack targets.

Deduction: explanation stability is useful for trust, but in cybersecurity it must be treated as a possible attack surface.

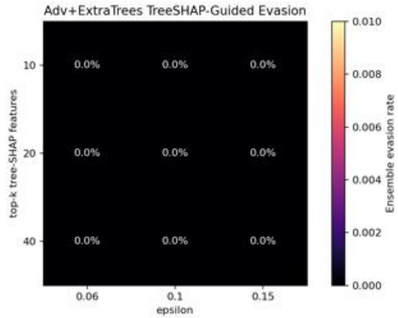
Explanations Help Analysts, But Also Guide Evasion



RF SHAP attack: best local evasion reaches 18.22%.



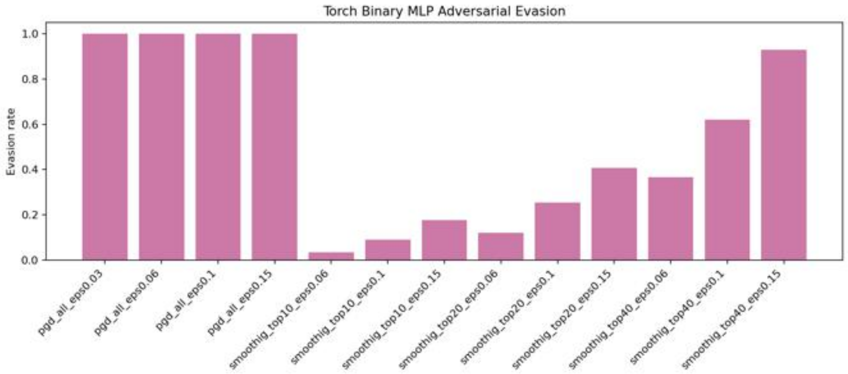
IG attack: best evasion reaches 15.97% on the Torch MLP.



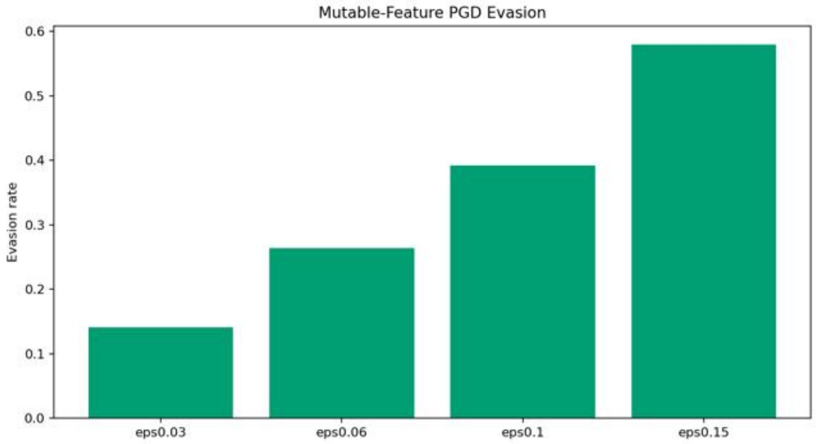
Final ensemble: TreeSHAP-guided evasion is 0.00% under

Security implication: explainability must be evaluated as an attack surface, not only as a transparency tool.

Neural IDS Alone Remains Vulnerable



Full-feature PGD: a white-box stress test reaches 100% evasion on the original Torch Binary MLP.



Mutable-feature PGD after fine-tuning: categorical/binary fields are frozen; strongest

Adversarial fine-tuning helps: mutable PGD evasion drops by 32.03 pp at eps 0.10 and 42.03 pp at eps 0.15.

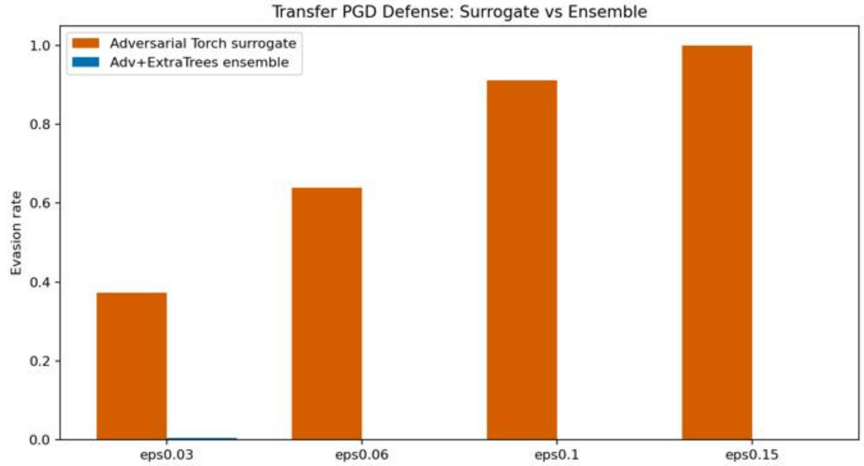
Limit: high-epsilon direct white-box PGD remains strong, so the neural detector should not be deployed alone.

The Final Ensemble Breaks Transferability

Transfer-PGD defense table

eps	Surrogate	Ensemble	Reduction
0.03	37.29%	0.50%	36.79 pp
0.06	63.99%	0.00%	63.99 pp
0.10	91.24%	0.00%	91.24 pp
0.15	100.00%	0.00%	100.00 pp

Empirical robustness under the tested transfer attack, not certified robustness.



Adversarial examples that fool the Torch surrogate transfer poorly to the robust tree-heavy ensemble.

Main defense deduction: adversarial fine-tuning helps the neural part, while SHAP-augmented ExtraTrees and ensembling reduce transferability.

Security-OR Is Optional, Not The Main Claim

Default model

0.9063
Adv+ExtraTrees F1

0.9064
Balanced accuracy

Chosen because it gives the best clean operating balance while keeping strong transfer-PGD defense.

Strict mode

0.8796
Security-OR F1

0.00%
Transfer evasion

Useful when false negatives are more costly than extra alerts, but not the default because clean F1 is lower.

Limits to state: NSL-KDD is old, feature-space attacks are imperfect, R2L/U2R remain difficult, and robustness is empirical rather than certified.

What This Project Demonstrates

0.9063
Final binary F1

0.845
Final local explanation stability

0.00%
Transfer evasion at eps >= 0.06

0.806
U2R recall

Contribution

One reproducible CUDA pipeline connects IDS performance, explainability, stability, adversarial attacks, and defense evaluation.

Final message

Explainability improves analyst trust, but in cybersecurity it must also be tested as a possible attack guide.

Thank you.